CRISPR gRNA Multi-Objective Optimization Framework

Mathematical Foundations, Biophysical Backend Algorithms, & Convex Optimization Architecture

Document Scope: Comprehensive Mathematical & Algorithmic Backend Specification • **Locus:** Human TP53 (GRCh38.p14) • **Population:** N = 2,860 Candidates • **Solver:** CVXOPT Convex Linear Program

1. Mathematical Problem Formulation & Global Architecture

The multi-objective CRISPR guide RNA (gRNA) optimization framework transforms a continuous genomic locus into a globally optimal subset of guide RNAs. The computational backend is rigorously structured as a **Multi-Objective Constrained Linear Program (LP)** operating over a convex feasible polytope. Candidate protospacers are simultaneously evaluated on on-target cleavage potency (Efficiency), genome-wide non-specific cleavage risk (Specificity), and biophysical thermodynamics (Structural Suitability).

The complete transformation maps nucleotide sequences across continuous mathematical spaces as follows:

```
Sequence Space Sigma^20 --[Feature Extraction]--> Feature Space R^23 --[Normalization]--> Quality Domain [0, 1]^23 --[Weights]--> Objective Vector c in [0, 1]^N --[Constraints]--> Feasible Polytope F in [0, 1]^N --[CVXOPT LP]--> Optimal Solution x* in {0, 1}^N
```

Pipeline Stage	Mathematical Mapping	Algorithmic Transformation	Domain & Dimensions
1. Protospacer Scan	`s_i in {A,C,G,T}^20`	Regular expression detection: 5'-N20-NGG-3' on forward/reverse strands	N = 2,860 candidates
2. Feature Extraction	`Phi: Sigma^20 -> R^23`	Thermodynamic NN kinetics, ViennaRNA MFE, Bowtie BWT alignment	23 features / guide
3. Normalization	`N: R -> [0, 1]`	Direction-aware Min-Max rescaling & Derringer-Suich desirability	E_i, S_i, R_i in [0, 1]
4. Objective Score	`c_i = w^T f_norm_i`	Weighted linear aggregation: c_i = 0.35 E_i + 0.40 S_i + 0.25 R_i	c_i in [0, 1]
5. Feasibility Bound	`u_i in {0, 1}`	Biological indicator: GC in [40,60]%, Tm in [55,65] deg C, Off-targets <= 20	F subset of R^2860 (5 pass)
6. CVXOPT Solver	`max c^T x s.t. x in F`	Primal-dual interior-point method solving bounded LP under budget K=10	x* = [1, 1, 1, 1, 1]^T

2. Thermodynamic & Biophysical Chemistry Calculations

A. SantaLucia & Hicks (2004) Nearest-Neighbor Duplex Kinetics: The thermodynamic stability of the 20-nt gRNA:DNA heteroduplex is computed via nearest-neighbor (NN) stacking interactions. Total enthalpy (dH) and entropy (dS) are computed as the sum of helix initiation, terminal A/T penalties, and 19 consecutive dinucleotide stacking steps:

```
dH = dH_init + dH_term(5') + dH_term(3') + sum_{j=1}^{19} dH_stack(s_j s_{j+1} / s_bar_j s_bar_{j+1})
dS = dS_init + dS_term(5') + dS_term(3') + sum_{j=1}^{19} dS_stack(s_j s_{j+1} / s_bar_j s_bar_{j+1})
```

The Gibbs Free Energy of Duplex Formation at physiological temperature (T = 37 deg C = 310.15 K) and the Nearest-Neighbor Melting Temperature (Tm) are determined via the standard van 't Hoff relation:

```
dG_37 = dH - T * dS (kcal/mol)
Tm = [ dH / ( dS + R * ln(C_T / 4) ) ] - 273.15 (deg C)
where R = 1.9872 cal/(mol*K) is the universal gas constant, and C_T = 250 * 10^-9 M is the total oligonucleotide concentration.
```

B. ViennaRNA Secondary Structure Minimum Free Energy (MFE): Self-folding of the single-stranded gRNA spacer can inhibit Cas9 ribonucleoprotein complex assembly. Secondary structure folding is modeled using dynamic programming (Zuker algorithm) via ViennaRNA. The Minimum Free Energy (dG_MFE) minimizes loop decomposition energy:

```
dG_MFE(s) = min_{S in ValidStructures(s)} [ sum_{h in Hairpins} dG_H(h) + sum_{i in Interior} dG_I(i) + sum_{b in Bulges} dG_B(b) + sum_{m in Multi} dG_M(m) ]
Paired Base Fraction: f_paired = (1 / L) * sum_{k=1}^L I(S_k in {'(', ')'})
```

3. Specificity Modeling, Burrows-Wheeler Transform & Mismatch Algebra

A. Bowtie 1.3.1 Genome-Wide Indexing: All 2,860 candidate sequences are aligned against the entire human reference genome (GRCh38.p14) using the Burrows-Wheeler Transform (BWT) and FM-Index with up to 3 mismatches (-v 3 -k 1000). Exactly one 0-mismatch alignment corresponds to the intended on-target locus and is deducted from the off-target tally:

```
OT_0 = max(Hits_0mm - 1, 0), OT_1 = Hits_1mm, OT_2 = Hits_2mm, OT_3 = Hits_3mm
Total Off-Target Burden: OT_total = OT_0 + OT_1 + OT_2 + OT_3
```

B. Mismatch Divergence & Capping Handling: To account for target discrimination, the mean edit distance across off-targets is computed. Highly divergent off-targets (3 mismatches) represent lower cleavage risk than near-perfect matches (1 mismatch):

```
d_mean_mismatch = ( sum_{m=0}^3 m * OT_m ) / ( sum_{m=0}^3 OT_m ) = ( 0*OT_0 + 1*OT_1 + 2*OT_2 + 3*OT_3 ) / OT_total
Zero Off-Target Imputation: if OT_total = 0 => d_mean_mismatch = max_j(d_mean_mismatch_j) (Safest bound)
Alignment Capping Penalty: if OT_total >= 999 => OT_norm = 0.0 (Worst-burden floor for truncated searches)
```

4. Multi-Objective Normalization & Composite Scoring Algebra

Individual biological features exist across vastly different dimensional scales (deg C, kcal/mol, hit counts). To prevent scale distortion, features undergo direction-aware linear min-max normalization and non-linear desirability transforms:

```
Direction-Aware Linear Scaling: N_max(X) = (X - X_min) / (X_max - X_min), N_min(X) = 1.0 - (X - X_min) / (X_max - X_min)

Derringer-Suich Trapezoidal Plateau Desirability for Tm (Optimal Range [55 deg C - 65 deg C], tolerance delta = 5 deg C):
d(Tm) = 1.0 if 55 <= Tm <= 65; (Tm - 50)/5 if 50 <= Tm < 55; (70 - Tm)/5 if 65 < Tm <= 70; 0.0 otherwise
```

Three Aggregated Sub-Objective Scores:

```
1. Efficiency (E_i): E_i = N_max(Doench_Proxy_i)
2. Specificity (S_i): S_i = 0.70 * N_min(OT_total_i) + 0.30 * N_max(d_mean_mismatch_i)
3. Structural Suitability (R_i): R_i = (1/3) * [ d(Tm_i) + N_min(dG_i) + N_max(dG_MFE_i) ]
```

Composite Multi-Objective Quality Function (c_i):

```
c_i = w_E * E_i + w_S * S_i + w_R * R_i = 0.35 * E_i + 0.40 * S_i + 0.25 * R_i (where sum w = 1.0)
```

5. Constrained Convex Optimization, Linear Programming & CVXOPT Solver

A. Core Optimization Principle & Distinct Mathematical Roles:
It is mathematically crucial to distinguish the three fundamental pillars of the system:
• **Weighted Scoring Function (c_i):** Evaluates the multi-objective utility of candidate guides.
• **Biological Constraints (u_i):** Delineates the bounding hyperplanes of the feasible convex polytope F.
• **CVXOPT Linear Program:** Solves the resource-constrained optimization problem to find the global optimum over F.

B. Biological Feasibility Indicator Vector (u in {0, 1}^N):
A guide RNA candidate i is biologically feasible (u_i = 1) if and only if it satisfies all 4 experimental bounds simultaneously:

```
u_i = I(0.40 <= GC_i <= 0.60) * I(55.0 deg C <= Tm_i <= 65.0 deg C) * I(OT_total_i <= 20) * I(Capped_i == False)
```

C. Primal Linear Program Formulation:

```
maximize sum_{i=1}^N c_i * x_i = c^T x
subject to 0 <= x_i <= u_i forall i in {1, ..., N} (Box constraints bounded by feasibility)
sum_{i=1}^N x_i <= K (Selection capacity budget constraint; K = 10)
```

Formal Mathematical Proofs of Convexity & Binary Vertex Optimality

Proof 1 (Convexity of Objective Function): The objective function f(x) = c^T x is a linear map from R^N to R. For any x, y in R^N and alpha in [0, 1]:
f(alpha * x + (1 - alpha) * y) = c^T(alpha * x + (1 - alpha) * y) = alpha * c^T x + (1 - alpha) * c^T y = alpha * f(x) + (1 - alpha) * f(y)
Because equality holds, f(x) is both convex and concave (affine). Maximizing an affine function over a convex set is a canonical convex optimization problem.

Proof 2 (Convexity of the Feasible Set F): The feasible set F is defined by 2N + 1 linear inequalities:
F = { x in R^N | -x_i <= 0, x_i <= u_i, sum_{i=1}^N x_i <= K }
Each linear inequality a_j^T x <= b_j defines a closed halfspace H_j in R^N. Halfspaces are convex sets. The feasible region is the intersection F = intersect_{j=1}^{2N+1} H_j. Since the intersection of any collection of convex sets is convex, F is a bounded convex polytope.

Proof 3 (LP Vertex Extreme Point & Binary Integrality Exactness): By the Fundamental Theorem of Linear Programming, if a linear program has an optimal solution, it is attained at an extreme point (vertex) of the polytope F. The constraint matrix G is totally unimodular. Therefore, all basic feasible solutions (vertices) are integral: x_vertex in {0, 1}^N. The continuous LP relaxation solved by CVXOPT guarantees the exact combinatorial integer optimum without branch-and-bound relaxation loss.

CVXOPT Standard Matrix Formulation (Primal-Dual Form)

CVXOPT solves convex optimization problems in standard minimization format:
minimize q^T x **subject to** G x <= h
To maximize c^T x, we set q = -c in R^N. The constraint system G x <= h is constructed via block matrices:

```
G = [ -I_{N x N} ; I_{N x N} ; 1_{1 x N} ] in R^{(2N+1) x N}
h = [ 0_{N x 1} ; u_{N x 1} ; K ] in R^{2N+1}
```

Karush-Kuhn-Tucker (KKT) Optimality Conditions:
The primal-dual interior-point method introduces Lagrange multipliers lambda >= 0 (lower bounds), mu >= 0 (upper bounds), and gamma >= 0 (budget constraint):

```
1. Primal Feasibility: -x_i <= 0, x_i - u_i <= 0, sum x_i - K <= 0
2. Dual Feasibility: lambda_i >= 0, mu_i >= 0, gamma >= 0
3. Stationarity: grad_x L = -c - lambda + mu + gamma * 1 = 0 => c_i = mu_i - lambda_i + gamma
4. Complementary Slackness: lambda_i * x_i = 0, mu_i * (u_i - x_i) = 0, gamma * (K - sum x_i) = 0 (forall i)
```

6. Empirical Dataset Results & Complete Verification (Human TP53)

The multi-objective optimization framework was executed on the complete human **TP53** gene dataset (20,394 bp exonic/intronic region). A total of **2,860** unique 20-nt protospacers upstream of NGG PAM motifs were identified and evaluated genome-wide against **GRCh38.p14**.

Reduction Step	Filter Bound Applied	Surviving Candidates	Retention %	Mathematical Impact on Feasible Set
Step 0: Raw Candidates	All NGG PAM upstream 20-nt sites	**2,860**	100.0%	Unconstrained search space R^2860
Step 1: GC Content	`0.40 <= GC <= 0.60`	**1,708**	59.7%	Eliminates unstable binding / extreme secondary folding
Step 2: Melting Temp (Tm)	`55.0 deg C <= Tm <= 65.0 deg C`	**991**	34.7%	Restricts search to physiological annealing temperatures
Step 3: Off-Target Ceiling	`Off-Target Count <= 20 hits`	**15**	0.52%	Eliminates severe genome-wide promiscuous cleavage risk
Step 4: All Feasible Bounds	`GC & Tm & OT <= 20 & Uncapped`	**5**	0.17%	Feasible Polytope F bounded to 5 candidate guides
Step 5: CVXOPT Optimal LP	`max c^T x s.t. x in F, sum x <= 10`	**5 Selected**	0.17%	Optimal vertex attainment with binary LP decisions x_i* = 1.0

Optimal CVXOPT-Selected Guide RNAs (Complete Solution Vector)

Rank	Guide Sequence (5' -> 3')	PAM	c_i	E_i (35%)	S_i (40%)	R_i (25%)	GC%	Tm (deg C)	OT	x_i*
#1	`GTACGCTTTAGGCCGGCTCA`	AGG	**0.8681**	0.732	1.000	0.848	60.0%	56.72	9	**1.000**
#2	`GCTGAACGTCGTGAAGCGGA`	AGG	**0.8554**	0.732	0.998	0.800	60.0%	57.61	12	**1.000**
#3	`ACGCTACCATGGCGTGAGAC`	AGG	**0.8385**	0.732	0.996	0.736	60.0%	56.74	15	**1.000**
#4	`CTGAACGTCGTGAAGCGGAA`	GGG	**0.7979**	0.557	0.989	0.830	55.0%	55.21	15	**1.000**
#5	`TAGGCATCACGGAGCGGTTA`	GGG	**0.7934**	0.557	0.977	0.831	55.0%	55.10	17	**1.000**

Comparison: Unconstrained Ranking vs. Constrained CVXOPT Optimization

A critical mathematical insight is why **unconstrained ranking fails**. For example, unconstrained Rank #1 (GAGGACCGTCGCAATCTGAG) achieves an unconstrained score of c_i = 0.9354. However, it exhibits a non-physiological melting temperature (Tm = 54.78 deg C < 55 deg C) and excessive off-target hits (OT = 23 > 20). Unconstrained ranking selects dangerous or non-functional guides, whereas CVXOPT strictly optimizes within the feasible region:

Guide Sequence	W-Rank	Objective c_i	Feasible?	Biological Violation Reason	CVXOPT LP Status
`GAGGACCGTCGCAATCTGAG`	#1	**0.9354**	**NO**	Tm = 54.78 deg C (bound [55-65]); Off-targets = 23 (ceiling <= 20)	**Rejected (u_i = 0)**
`GCCCGAACGCAAAGTGTCCC`	#2	**0.9330**	**NO**	GC content = 65.0% (allowed [40-60%])	**Rejected (u_i = 0)**
`GCAGGCTGAACGTCGTGAAG`	#3	**0.9284**	**NO**	Off-target count = 38 hits (ceiling <= 20)	**Rejected (u_i = 0)**
`GTACGCTTTAGGCCGGCTCA`	#40	**0.8681**	**YES**	Satisfies all 4 bounds (GC=60%, Tm=56.72 deg C, OT=9)	**Selected (Rank #1, x_i*=1)**
`GCTGAACGTCGTGAAGCGGA`	#58	**0.8554**	**YES**	Satisfies all 4 bounds (GC=60%, Tm=57.61 deg C, OT=12)	**Selected (Rank #2, x_i*=1)**

7. Master Mathematical Notation & Algorithmic Reference

Symbol	Mathematical Concept	Exact Formula / Algorithmic Definition	Biological & Optimization Role
s_i	Guide RNA Protospacer	s_i in {A, C, G, T}^20	20-nt target sequence upstream of 5'-NGG-3' PAM
dG_37	Duplex Free Energy	dH - 310.15 * dS (SantaLucia 2004 NN)	Thermodynamic hybridization binding stability
Tm	Melting Temperature	dH / (dS + R * ln(C_T/4)) - 273.15	Physiological annealing threshold (optimal 55-65 deg C)
dG_MFE	Folding Free Energy	min_S sum dG_loop (ViennaRNA dynamic prog.)	Penalizes stable self-folding loops / hairpins
OT_total	Off-Target Burden	sum_{m=0}^3 Hits_m - 1 (Bowtie GRCh38)	Total off-target cleavage count across human genome
d_mismatch	Mismatch Divergence	(sum m * OT_m) / OT_total	Mean edit distance across off-target sites
E_i	Normalized Efficiency	N_max(Doench_Proxy_i) in [0, 1]	On-target cleavage potency proxy (35% weight)
S_i	Normalized Specificity	0.70 N_min(OT_i) + 0.30 N_max(d_mismatch_i)	Safety / off-target avoidance score (40% weight)
R_i	Structural Suitability	(1/3) [d(Tm_i) + N_min(dG_i) + N_max(dG_MFE_i)]	Thermodynamic stability composite (25% weight)
c_i	Objective Coefficient	0.35 E_i + 0.40 S_i + 0.25 R_i	Guide quality scalar utility in linear program
u_i	Feasibility Indicator	I(GC in [40,60]%) * I(Tm in [55,65]) * I(OT <= 20)	Delineates bounding hyperplanes of polytope F
x_i	Decision Variable	x_i in [0, u_i] subset of [0, 1]	Continuous LP selection variable (attains binary {0,1})
K	Selection Budget	sum_{i=1}^N x_i <= K (K = 10)	Upper bound on total guides selected for synthesis
CVXOPT	LP Interior Solver	cvxopt.solvers.lp(q, G, h)	Primal-dual interior-point method finding global optimum